
Transduction - High Quality

Introduction

This protocol describes how to perform a retroviral transduction.

Materials

› Materials

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Procedure

Day 1

1. Cd8 T-Cell Isolation and Activation.
2. Get 6 well non-TC plate. Giovanni hides them above the cell counter in hedrick TC. Ask Natalie for the extras.
3. Make the following mix. It is recommended to make a bit more than you actually need so add 1 to the number of wells you have.

Retronectin mix				^
	Name of reagent	Amount:	Number of wells + 1	
1	DPBS	6205.5	6.3	
2	Retronectin	94.5		

4. 1 ml of mix per well in the special tc plate.
- 5.
6.  **CRITICAL** wrap the plate in parafen. BE VERY CAREFUL NOT TO SPLASH
7. Put in 4 degree cold room in giovanni's secret drawer.

Day 2

8. Put TCM in the hot water bath.
9. Remove the retronectin from each well in the coated plates.

10. Add 4 ml of the warmed TCM drop-by-drop to each well to block aspecific sites.

11. Incubate at 37 C for 30 min.

00:30:00



 **CRITICAL** Grab all the viruses that you want to use from the -80 and thaw them in a ziploc bag in a hot water bath. As soon as they are fully thawed put them on ice.

12. Remove TCM and add 750 ul of virus to respective wells.

13. Incubate at 37 C for 30 min.

00:30:00



14. Remove the virus and add new 750 ul of virus to each well.

15. Incubate at 37 C for 30 min.

00:30:00



16.  **CRITICAL** Prepare cytokines with 10 min left on previous step

Cytokine Mix					^
	A	B	C	D	
1					

17. Pool together cells -spin down for 1600, 4 min roomt emp

18. Prepare 2 ml of TCM + IL2 --> add 1:1667 IL2 (1.2 ul)

19. 500 ul to each well

20. Add 1.5 ml of virus to each well

21. 1600 rpm, 37 C,. 40 min of the 6Wp

22. Incubate at 37 C for 24 hr

23. Transfer recipients to pac

Day 3

24. Grab transduced cells 6 well plate from the incubator.
25. Collect the different cell population separately into 15 ml conicals. Use 1 ml of MACS buffer to help wash the cells of the wells.
26. Top up the conicals with PBS
27. Centrifuge at 1600 rpm for 4 minutes.

Filtering

28. Resuspend the conical in 300 ul MACS buffer.
29. Filter cells through those little white square sheets in the TC hood.
30. Prepare collection tubes

Sterile FACS tubes with 500 ul of FBS

Sorting

31. Sort cells - no idea how to do this
- 32.
33.  **CRITICAL** Wash cells after sorting. Aspirate the liquid using the vacuum. Be extremely careful not to vacuum up the cells
34. Resuspend in 500 ul PBS
35. Count cells 10 ul: 10ul. Do the math to figure out how to mix and transfer the cells.

Prepare LCMV

Injection